STD 2: Measurement (Std 2), M7 Rates and Ratios (Y12)

Ratio and Scale (Std2)

Teacher: Robert Tram

Exam Equivalent Time: 102 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-M7 Rates and Ratios* is a new syllabus category for a wide range of previous Gen2 content such as best buys, medication, energy consumption, ratio and scale questions.
- This analysis will look at *Ratio and Scale.*

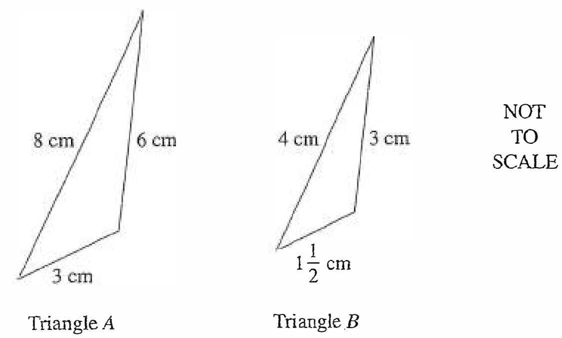
ANALYSIS - What to Expect and Common pitfalls

- *Ratio and Scale* has surprised in the first two Std2 exams, with allocations of 5 marks and 7 marks. Although early days, any revision planning should reflect that this topic area has been attracting meaningfully higher mark allocations than in the past.
- *Ratios* were examined in 2020 and 2019 with a new focus and question style that warrants attention. Any revision must include 3-part ratios, as required in both years.
- Prior to 2019, *ratios* have typically been examined through house plans and maps (to mixed results), with the notable exception of *2013 Std2 30c*, where a fuel mixing context caused massive problems and should be carefully reviewed.
- *Similarity* has been examined in past Std2 exams and can arguably be included within the syllabus dot point *"solve practical problems using ratio"*. We include a number of these examples in the database.
- Capture/Recapture questions are now confirmed as examinable content (NESA Nov-19 syllabus maintenance). The database includes numerous examples that produced sub-50% mean marks to cover this area.
- *Scale* is a core aspect of ratios and was tested in 2019 in a challenging cross topic question with the trapezoidal rule (*2019 Std2 41*).

Questions

1. Measurement, STD2 M7 2007 HSC 4 MC

What scale factor has been used to transform Triangle A to Triangle B?



(A) $\frac{1}{2}$

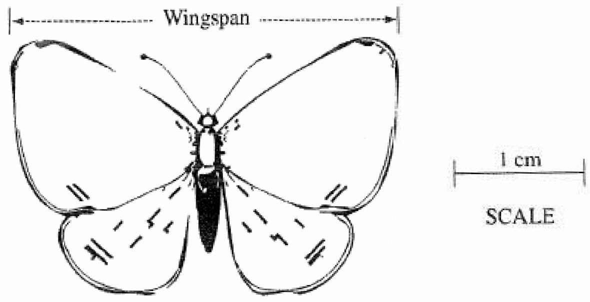
(B) $\frac{3}{4}$

(C) 2

(D) 3

2. Measurement, STD2 M7 2005 HSC 4 MC

The diagram is a scale drawing of a butterfly.



What is the actual wingspan of the butterfly?

(A) 2.5 cm

(B) 3 cm

(C) 15 cm

(D) 8.75 cm

3. Measurement, STD2 M7 SM-Bank 3 MC

There are 8 male chimpanzees in a community of 24 chimpanzees.

What is the ratio of males to females in the community?

A. 1 : 3

B. 1 : 2

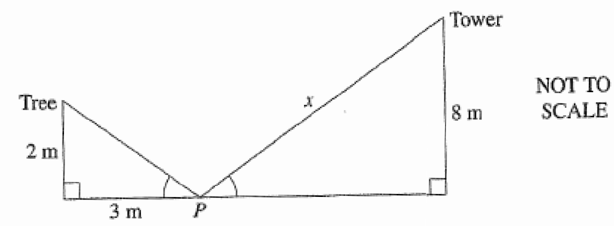
C. 3 : 1

D. 2 : 1

4. Measurement, STD2 M7 2008 HSC 20 MC

A point P lies between a tree, 2 metres high, and a tower, 8 metres high. P is 3 metres away from the base of the tree.

From P, the angles of elevation to the top of the tree and to the top of the tower are equal.



What is the distance, x, from P to the top of the tower?

(A) 9 m

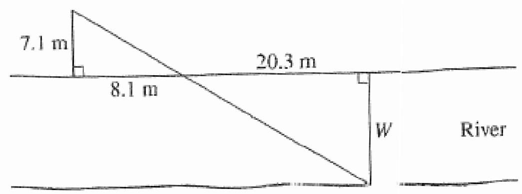
(B) 9.61 m

(C) 12.04 m

(D) 14.42 m

5. Measurement, STD2 M7 2016 HSC 16 MC

The width (W) of a river can be calculated using two similar triangles, as shown in the diagram.



What is the approximate width of the river?

(A) 17.8 m

(B) 19.3 m

(C) 23.2 m

(D) 24.9 m

6. Measurement, STD2 M7 SM-Bank 1 MC

A dog breeder owns 500 dogs.

150 are Ridgebacks and the rest are Pugs.

What is the ratio of Ridgebacks to Pugs?

A. 3 : 5

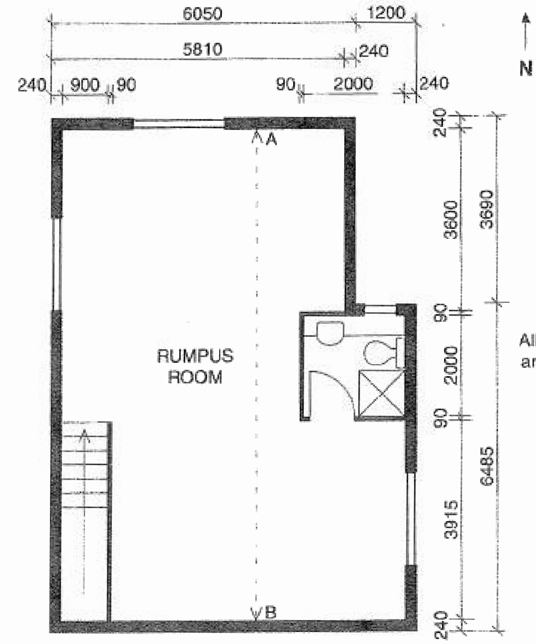
B. 3 : 7

C. 3 : 9

D. 3 : 10

7. Measurement, STD2 M7 2011 HSC 24a

Part of the floor plan of a house is shown. The plan is drawn to scale.



All measurements are in millimetres.

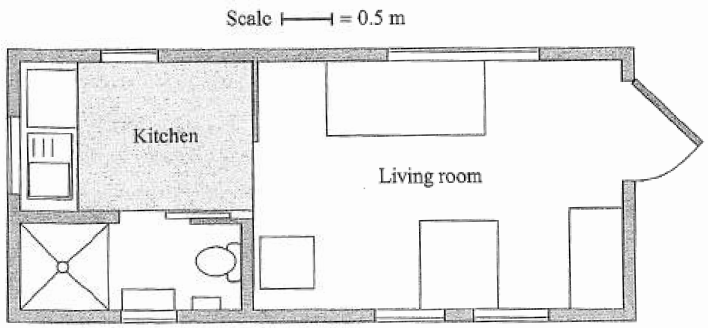
i. What is the width of the stairwell, in millimetres? *(1 mark)*

ii. What are the internal dimensions of the bathroom, in millimetres? *(1 mark)*

iii. What is the length AB, the internal length of the rumpus room, in millimetres? *(1 mark)*

8. Measurement, STD1 M5 2019 HSC 20

The plan of the lower level of a small house is shown.



a. How many windows are shown on the plan? *(1 mark)*

b. What is the actual perimeter, in metres, of the shaded part of the kitchen floor? *(2 marks)*

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9. Measurement, STD2 M7 2019 HSC 18

Andrew, Brandon and Cosmo are the first three batters in the school cricket team. In a recent match, Andrew scored 30 runs, Brandon scored 25 runs and Cosmo scored 40 runs.

a. What is the ratio of Andrew's to Brandon's to Cosmo's runs scored, in simplest form? *(2 marks)*

b. In this match, the ratio of the total number of runs scored by Andrew, Brandon and Cosmo to the total number of runs scored by the whole team is $19:36$.
How many runs were scored by the whole team? *(2 marks)*

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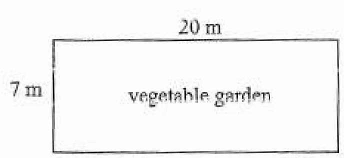
10. Measurement, STD2 M7 SM-Bank 10

A potting mix is made up of two ingredients, potting soil and manure, in the ratio 5:2.

21 kilograms of potting mix are used to fertilise the rectangular vegetable garden pictured below.



i. What rate must the whole 21 kilograms of potting mix be applied to be spread evenly over the garden. Give your answer in grams per square metre. *(2 marks)*

ii. How much manure is needed to make the required potting mix? *(1 mark)*

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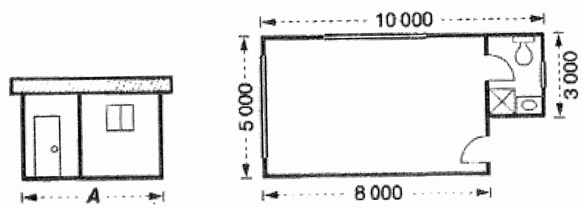
11. Measurement, STD2 M7 2010 HSC 23b

The elevation and floor plan of a building are shown.



All measurements are in millimetres.

Calculate the area of the floor of this building in square metres. *(2 marks)*

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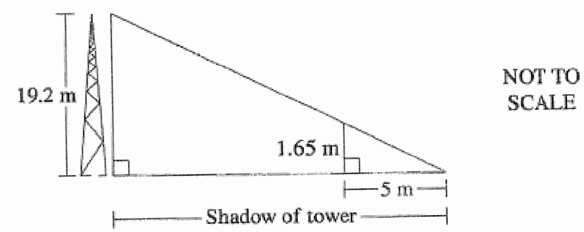
12. Measurement, STD2 M7 2015 HSC 27a

At a particular time during the day, a tower of height 19.2 metres casts a shadow. At the same time, a person who is 1.65 metres tall casts a shadow 5 metres long.



What is the length of the shadow cast by the tower at that time? *(2 marks)*

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13. Measurement, STD2 M7 SM-Bank 6

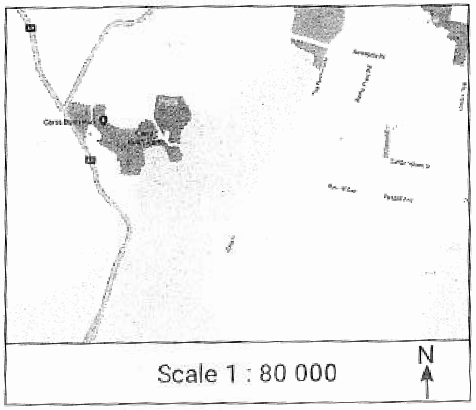
In a raffle, the total prize money is shared among the first two tickets drawn in the ratio 5 : 4.

The prize for the second ticket drawn is $360.

What is the total prize money? *(2 marks)*

14. Measurement, STD2 M7 SM-Bank 7

Part of a map is shown.



The distance between two points on the above map (not shown) is 3.4 kilometres. How far apart on the map would be the two points be, in centimetres? ***(2 marks)***

15. Measurement, STD2 M7 SM-Bank 13

A scalene triangle has angles in the ratio 9 : 2 : 4.

In degrees, what is the size of its largest angle? ***(2 marks)***

16. Measurement, STD2 M7 SM-Bank 17

There are 48 people in a meeting room.

The ratio of females to males in the room is 13 to 11.

How many females need to leave the room so that the ratio of females to males in the room is 1 to 1? ***(2 marks)***

17. Measurement, STD2 M7 2007 HSC 23c

A scientific study uses the 'capture-recapture' technique.

In the first stage of the study, 24 crocodiles were caught, tagged and released.

Later, in the second stage of the study, some crocodiles were captured from the same area. Eighteen of these were found to be tagged, which was 40% of the total captured during the second stage.

i. How many crocodiles were captured in total during the second stage of the study? ***(1 mark)***

ii. Calculate the estimate for the total population of crocodiles in this area. ***(2 marks)***

18. Measurement, STD2 M7 2016 HSC 28a

Jacob has a large jar of silver coins. He adds 20 gold coins into the jar. He then seals the jar and shakes it to ensure that the gold coins are mixed in thoroughly with the silver coins. Jacob then opens the jar and takes a handful of coins. In his hand he has 33 silver coins and 4 gold coins.

i. Based on Jacob's handful, if a coin is selected at random from the jar, what is the probability that it is a gold coin? ***(1 mark)***

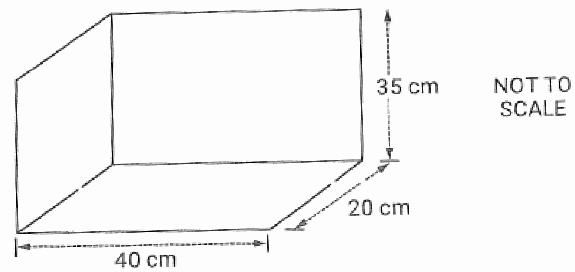
ii. Jacob returns the handful of coins to the jar. Estimate the total number of coins in the jar. ***(2 marks)***

19. Measurement, STD2 M7 2020 HSC 23

In a tropical drink, the ratio of pineapple juice to mango juice to orange juice is 15 : 9 : 4.

a. How much orange juice is needed if the tropical drink is to contain 3 litres of pineapple juice? ***(2 marks)***

b. The internal dimensions of a drink container, in the shape of a rectangular prism, are shown.



To completely fill the container with the tropical drink, how many litres of mango juice are required. ***(3 marks)***

20. Measurement, STD2 M7 2012 HSC 26f

The capture-recapture technique was used to estimate a population of seals in 2012.

- 60 seals were caught, tagged and released.
- Later, 120 seals were caught at random.
- 30 of these 120 seals had been tagged.

The estimated population of seals in 2012 was 11% less than the estimated population for 2008.

What was the estimated population for 2008? ***(2 marks)***

21. Measurement, STD2 M7 2012 HSC 27c

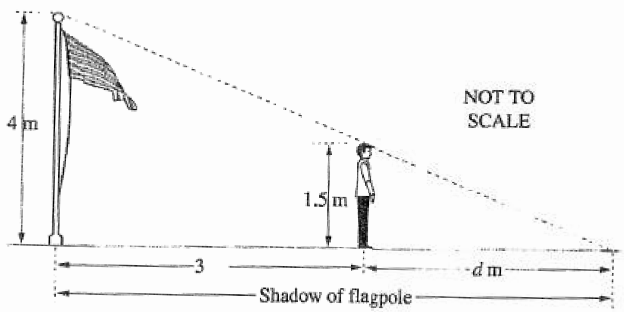
A map has a scale of 1 : 500 000.

i. Two mountain peaks are 2 cm apart on the map.
What is the actual distance between the two mountain peaks, in kilometres? *(1 mark)*

ii. Two cities are 75 km apart. How far apart are the two cities on the map, in centimetres? *(1 mark)*

22. Measurement, STD2 M7 2012 HSC 28c

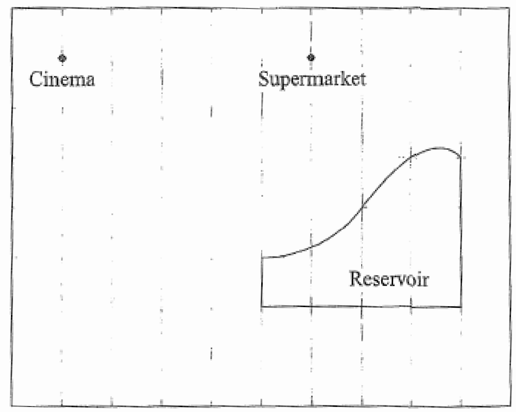
Jacques and a flagpole both cast shadows on the ground. The difference between the lengths of their shadows is 3 metres.



What is the value of d, the length of Jacques' shadow? *(3 marks)*

23. Measurement, STD2 M7 2019 HSC 41

A map is drawn to scale, on 1-cm paper, showing the position of a supermarket and a cinema. A reservoir is also shown.



a. It takes 10 minutes to walk in a straight line from the cinema to the supermarket at a constant speed of 3 km/h. Show that the scale of the map is 1 cm = 100 m. *(3 marks)*

b. The reservoir is initially empty. During a storm 20 mm of rain falls on the reservoir.
With the aid of one application of the trapezoidal rule, estimate the amount of water in the reservoir immediately after the storm. Assume that all rain which falls over the reservoir is stored. Give your answer in cubic metres. *(3 marks)*

24. Measurement, STD2 M7 SM-Bank 11

On a map, the distance between two towns is measured at 54 millimetres.

i. The actual distance between the two towns is 16.2 kilometres. What is the scale of the map? *(1 mark)*

ii. If two other towns on the same map are 9.2 centimetres apart, what is the actual distance between the two towns in kilometres? *(1 mark)*

25. Measurement, STD2 M7 SM-Bank 9

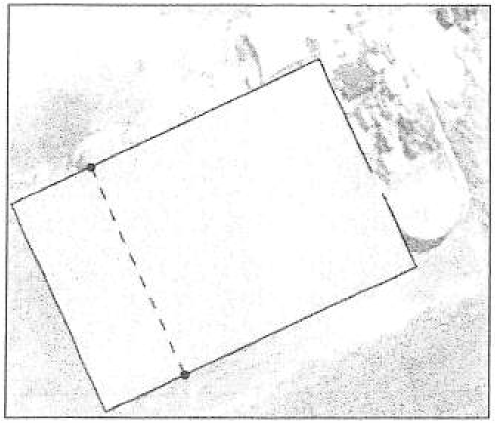
A farmer sells 216 sheep at a livestock auction.

Three buyers purchased all the sheep in the ratio 7:3:2

How many sheep did the buyer of the highest number of sheep purchase? *(2 marks)*

26. Measurement, STD2 M7 2015 HSC 29c

The image shows a rectangular farm shed with a flat roof.



The real width of the shed indicated by the dotted line was measured using an online ruler tool, and found to be approximately 12 metres.

i. By measurement and calculation, show that the area of the roof of the shed is approximately 216 m^2. *(2 marks)*

ii. All the rain that falls onto this roof is diverted into a cylindrical water tank which has a diameter of 3.6 m. During a storm, 5 mm of rain falls onto the roof.

Calculate the increase in the depth of water in the tank due to the rain that falls onto the roof during the storm. *(3 marks)*

27. Measurement, STD2 M7 2013 HSC 30c

Joel mixes petrol and oil in the ratio 40 : 1 to make fuel for his leaf blower.

i. Joel pours 5 litres of petrol into an empty container to make fuel for his leaf blower.

How much oil should he add to the petrol to ensure that the fuel is in the correct ratio? *(1 mark)*

ii. Joel has 4.1 litres of fuel left in his container after filling his leaf blower.

He wishes to use this fuel in his lawnmower. However, his lawnmower requires the petrol and oil to be mixed in the ratio 25 : 1.

How much oil should he add to the container so that the fuel is in the correct ratio for his lawnmower? *(3 marks)*

Worked Solutions

1. Measurement, STD2 M7 2007 HSC 4 MC

Take two corresponding sides

In ΔA 3 cm

In ΔB 1 ½ cm

$\therefore$ Scale factor converting ΔA to $\Delta B = ½$

$\Rightarrow A$

2. Measurement, STD2 M7 2005 HSC 4 MC

Wingspan is 3 times the scale distance that equals 1 cm.

$$\therefore \text{Wingspan} = 3 \times 1$$
$$= 3 \text{ cm}$$

$\Rightarrow B$

3. Measurement, STD2 M7 SM-Bank 3 MC

Males : Females

$8 : (24 - 8)$

$8 : 16$

$1 : 2$

$\Rightarrow$ B

4. Measurement, STD2 M7 2008 HSC 20 MC

Triangles are similar (equiangular)

In smaller triangle:

$$h^2 = 2^2 + 3^2$$
$$= 13$$
$$h = \sqrt{13}$$

$$\frac{x}{8} = \frac{\sqrt{13}}{2} \quad \text{(sides of similar } \Delta\text{s same ratio)}$$

$$x = \frac{8\sqrt{13}}{2}$$
$$= 14.422...$$

$\Rightarrow D$

5. Measurement, STD2 M7 2016 HSC 16 MC

Using similar ratios,

$$\frac{W}{7.1} = \frac{20.3}{8.1}$$
$$\therefore W = \frac{20.3 \times 7.1}{8.1}$$
$$= 17.79\ldots$$

$\Rightarrow A$

6. Measurement, STD2 M7 SM-Bank 1 MC

Since 150 are ridgebacks,

$\Rightarrow 500 - 150 = 350$ pugs

Ridgebacks : Pugs

$150:350$

$3:7$

$\Rightarrow B$

7. Measurement, STD2 M7 2011 HSC 24a

i. 900 mm

ii. 2000 mm $\times$ 2000 mm

iii. Length of Rumpus Room $= AB$

$$AB = 3600 + 90 + 2000 + 90 + 3915$$
$$= 9695 \text{ mm}$$

8. Measurement, STD1 M5 2019 HSC 20

a. 6 windows

b. Plan Perimeter $= 2 \times 3$ units $+ 2 \times 3.5$ units

$= 13$ units

$\therefore$ Actual perimeter $= 13 \times 0.5$

$= 6.5$ metres

9. Measurement, STD2 M7 2019 HSC 18

a. $A:B:C = 30:25:40$

$= 6:5:8$

b. Total runs by $A, B, C = 30 + 25 + 40$

$= 95$

Let $R =$ team runs

$$\frac{R}{95} = \frac{36}{19}$$
$$\therefore R = \frac{36 \times 95}{19}$$
$$= 180 \text{ runs}$$

10. Measurement, STD2 M7 SM-Bank 10

i. Potting mix: 21 kg $\Rightarrow$ 21, 000 grams

$$\text{Area of garden} = 20 \times 7 = 140 \text{ m}^2$$

$$\therefore \text{ Rate} = \frac{21\,000}{140} = 150 \text{ g/m}^2$$

ii. Ratio $5:2 \ \Rightarrow \ \frac{5}{7} : \frac{2}{7}$

$$\therefore \text{ Amount of manure} = \frac{2}{7} \times 21 = 6 \text{ kg}$$

11. Measurement, STD2 M7 2010 HSC 23b

MARKER'S COMMENT: Less errors were made by students who converted to metres *FIRST*, before calculating the area.

$$\text{Floor Area} = \text{Area 1} + \text{Area 2}$$
$$= (8 \times 5) + (2 \times 3) \quad (1 \text{ m} = 1000 \text{ mm})$$
$$= 46 \text{ m}^2$$

12. Measurement, STD2 M7 2015 HSC 27a

Both triangles have right angles and a common angle to the ground.

$\therefore$ Triangles are similar (equiangular)

Let $x =$ length of tower shadow

$$\frac{x}{19.2} = \frac{5}{1.65} \quad \text{(corresponding sides of similar triangles)}$$

$$x = \frac{5 \times 19.2}{1.65} = 58.1818\ldots = 58 \text{ m (nearest m)}$$

13. Measurement, STD2 M7 SM-Bank 6

$$\frac{4}{10} \times \text{Total prize money} = \$360$$
$$\Rightarrow \frac{1}{10} \times \text{Total prize money} = \$90$$
$$\therefore \text{ Total prize money} = 90 \times 10 = \$900$$

14. Measurement, STD2 M7 SM-Bank 7

Actual distance on map

$$= \frac{\text{real distance}}{\text{scale}} = \frac{3.4 \text{ km}}{80\,000} = \frac{3400 \text{ m}}{80\,000} = 0.0425 \text{ m} = 4.25 \text{ cm}$$

23. Measurement, STD2 M7 2019 HSC 41

a. $3 \text{ km/h} = 3000$ metres per 60 minutes

In 10 minutes:

♦ Mean mark part (a) 49%.

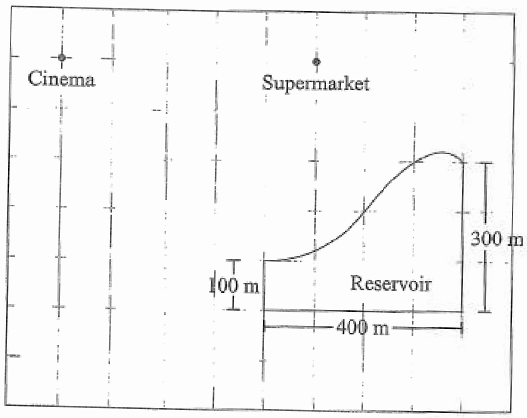
$$\text{Actual distance} = 3000 \times \frac{10}{60} = 500 \text{ metres}$$

Distance on map $= 5$ cm

$\therefore$ Scale 5 cm : 500 metres

1 cm : 100 metres

b.



$$A \approx \frac{h}{2}(a+b)$$

$$\approx \frac{400}{2}(100+300)$$

$$\approx 80\,000 \text{ m}^2$$

Converting mm to metres:

$$20 \text{ mm} = \frac{20}{1000} \text{m} = 0.02 \text{ metres}$$

♦♦ Mean mark part (b) 28%.

$\therefore$ Volume of water

$= A \times h$

$= 80\,000 \times 0.02$

$= 1600 \text{ m}^3$

24. Measurement, STD2 M7 SM-Bank 11

i. Convert both distance to the same unit (cm).

$54 \text{ mm} = 5.4 \text{ cm}$

$16.2 \text{ km} = 16\,200 \text{ m}$

$= 1\,620\,000 \text{ cm}$

$\therefore$ Scale $5.4 : 1\,620\,000$

$1 : 300\,000$

ii. Actual distance $= 9.2 \times 300\,000$

$= 2\,760\,000 \text{ cm}$

$= 27\,600 \text{ m}$

$= 27.6 \text{ km}$

25. Measurement, STD2 M7 SM-Bank 9

$$7:3:2 \Rightarrow \frac{7}{12} : \frac{3}{12} : \frac{2}{12}$$

$\therefore$ Buyer of the most sheep

$$= \frac{7}{12} \times 216$$

$= 126$ sheep

26. Measurement, STD2 M7 2015 HSC 29c

i. By measurement,

Roof length = 1.5 times the roof width

Width = 12 m (given)

$$\text{Length} = 1.5 \times 12 = 18 \text{ m}$$

$$\text{Area of roof} = 12 \times 18 = 216 \text{ m}^2 \text{ ...as required}$$

♦ Mean mark 35%.

ii. Volume of water

$$= Ah$$
$$= 216 \times 0.005 \quad (5 \text{ mm} = 0.005 \text{ m})$$
$$= 1.08 \text{ m}^3$$

♦♦♦ Mean mark 14%.

Volume of cylinder $= \pi r^2 h$

$$r = \frac{3.6}{2} = 1.8 \text{ m}$$

Find h when $V = 1.08 \text{ m}^3$,

$$\pi \times 1.8^2 \times h = 1.08$$

$$\therefore h = \frac{1.08}{\pi \times 1.8^2}$$
$$= 0.1061\ldots \text{ m}$$
$$= 10.6 \text{ cm (to 1 d.p.)}$$

27. Measurement, STD2 M7 2013 HSC 30c

i. Petrol : Oil = 40 : 1

♦ Mean mark 50%

5 Litres : X = 40 : 1

$$\therefore \frac{5}{X} = \frac{40}{1}$$

$$40X = 5$$

$$X = \frac{5}{40} = 0.125 \text{ L}$$

Joel should add 0.125 L of oil

ii. 4.1 L = 4100 mL

In ratio 40:1, there is

$\Rightarrow$ 4000 mL Petrol and 100 mL Oil

♦♦♦ Mean mark 8%
STRATEGY: The critical information required is how much ***petrol*** is in the container. Once known, the oil required to satisfy a 25:1 ration can be easily found.

Lawnmower ratio = 25 : 1

Let Oil required for 4000 mL Petrol = X mL

$$\frac{X}{4000} = \frac{1}{25}$$

$$25X = 4000$$

$$X = \frac{4000}{25} = 160 \text{ mL}$$

4 L of petrol requires 160 mL of oil

Container already has 100 mL of oil

$$\text{Oil to add} = 160 - 100 = 60 \text{ mL}$$